SP250 : Pascal's Triangle and Invariants

Pascal's Triangle

```
                1                ← row 0
              1   1
            1   2   1
          1   3   3   1
        1   4   6   4   1        ← row 4
      1   5  10  10   5   1
```

The kth number in row n is $\binom{n}{k}$.

- What is the sum of numbers in the nth row of the triangle?

```
                1                Sum = 1
              1   1                    2
            1   2   1                  4
          1   3   3   1                8
        1   4   6   4   1             16
```

Each number contributes twice to the next row.

What about the 1's on the edges?

They also contribute twice.

So:

$$\binom{n}{0} + \binom{n}{1} + \ldots + \binom{n}{n-1} + \binom{n}{n} = 2^n$$

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Remember Pascal's Identity (Rule)

$$\binom{0}{0}$$

$$\binom{1}{0}\ \binom{1}{1}$$

$$\binom{2}{0}\ \binom{2}{1}\ \binom{2}{2}$$

$$\binom{3}{0}\ \binom{3}{1}+\binom{3}{2}\ \binom{3}{3}$$

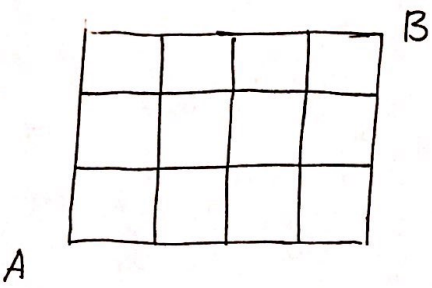
$$\binom{4}{0}\ \binom{4}{1}\ \binom{4}{2}\ \binom{4}{3}\ \binom{4}{4}$$

$$\rightarrow \binom{n}{k} = \binom{n-1}{k-1} + \binom{n-1}{k}$$

This is easy to see when you look at Pascal's Triangle.

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Problem 1



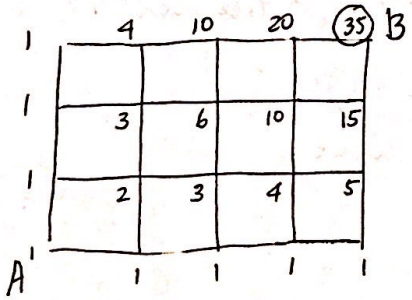
Mike walks from A to B along the grid.
He can only go up or to the right.
How many different paths can Mike take?

Solution

Mike must take exactly 7 steps. = 4 right + 3 steps up.

We choose 4 of the 7 steps to go right.

There are $\binom{7}{4} = 35$ paths.



This looks like Pascal's Triangle!

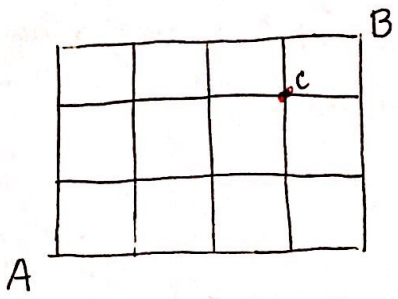
Note symmetry:

$$\binom{7}{4} = \binom{7}{3}$$

$$\binom{n}{k} = \binom{n}{n-k} \text{ for any } n, k.$$

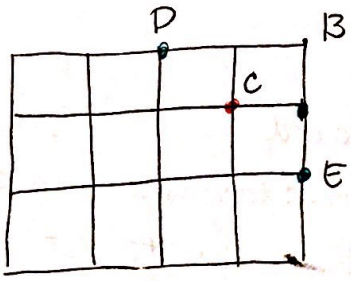
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Problem 2 : A variation on Problem 1:



Again Mike can only walk upwards and to the right to get from A to B. But now intersection C is blocked off. How many different paths can Mike take?

Solution 1 : direct



A valid path would go through D or E.

To D : $\binom{5}{2} = 10$ To E : $\binom{5}{1} = 5$

$10 + 5 =$ 15 paths total to B.

Solution 2 : Negative space

Calculate invalid paths A → C and C → B.

A to C : $\binom{5}{2} = 10$ C to B : $\binom{2}{1} = 2$

$10 \times 2 = 20$ invalid paths.

Originally we had 35 paths A → B.

So now with C blocked we have $35 - 20 =$ 15 paths.

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